

MUHAMMAD ARSLAN ANWER

Senior Full Stack Software Engineer · Node.js · NestJS · Express.js · Vue.js · React · TypeScript · AI

✉ arslan.anwer778@gmail.com | 📞 +92 302 7993900 | 📍 Lahore, Pakistan | 🔗 LinkedIn | 🐙 GitHub | 🌐 Portfolio

PROFESSIONAL SUMMARY

Results-driven **Full Stack Developer** with expertise in building scalable, secure, and high-performance web applications using **Node.js, NestJS, Express.js, React.js, Next.js, Vue.js, JavaScript, and TypeScript**. Experienced in designing RESTful APIs, microservices, authentication systems, and responsive user interfaces while integrating third-party services and optimizing application performance. Skilled in databases including **PostgreSQL, MySQL, MongoDB, and Redis**, with hands-on experience in **Docker, AWS, Git, CI/CD** and cloud deployment. Experienced in developing **AI-powered applications** by integrating Large Language Models (LLMs), OpenAI APIs, AI assistants, prompt engineering, Retrieval-Augmented Generation (RAG), and intelligent workflow automation. Passionate about building modern, reliable, and user-centric software solutions while following clean architecture and Agile development practices.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript
Backend	Node.js, NestJS, Express.js, REST API, WebSocket, Microservices, Bull Queue, node-cron, Swagger/OpenAPI
Frontend	Vue.js, React.js, Next.js, Nuxt.js
Databases	MySQL, PostgreSQL, MongoDB, Redis, Firebase Firestore · TypeORM, Prisma, Sequelize, Mongoose
AI Tools	Claude Code, OpenAI Codex, Cursor, GitHub Copilot, Warp AI, Windsurf, MCP Servers, LLM-Assisted Development
Tooling & APIs	Docker, Nginx, PM2, AWS S3, Firebase, CI/CD (GitHub, GitLab, Bitbucket), AI Integration, Prompt Engineering, Payment Gateways (Stripe, PayPal, 2Checkout, Cryptomus), Maps / GIS (Leaflet, Mapbox, MapLibre, Turf.js, Google Maps), Auth & Security

PROFESSIONAL EXPERIENCE

FAMS by Falkenherz (Voltro)
Senior Software Engineer – Full Stack

Lahore, Pakistan
February 2025 – Present

- Engineered a **multi-criteria Preventive Maintenance engine** (CAN bus engine hours, distance, period-based) with automated job card generation and missed-interval catch-up logic for a large-scale municipal fleet.
- Architected a **12-module Inventory Management system** with 5-stage PO approval workflow, rack-level batch stock tracking, consumption recording, and ExcelJS exports — replacing manual procurement processes.
- Built **real-time Leaflet.js geospatial dashboards** with Turf.js zone geofencing, GPS route playback, zone entry detection via ray-casting, and heatmap visualization for fleet operations.
- Contributed to **MM-UCCP smart waste platform**: IoT bin monitoring (fill-level, RFID, temp), sewage tanker dispatch with simulated annealing route optimization, alarm management, and multi-role architecture.
- Maintained **13 production cron jobs**; deployed on Oracle Cloud with Docker/Nginx, Node.js cluster mode (8 workers), and Bitbucket Pipelines CI/CD.

Stack: Node.js, Express.js, Vue 3, TypeScript, MySQL, Leaflet.js, MapboxGL, Turf.js, Docker, Nginx, Firebase FCM, Oracle Cloud

Septem Systems
Software Engineer – Full Stack Developer

Lahore, Pakistan
January 2022 – February 2025

- Delivered **8+ enterprise products end-to-end** across IoT, AI/ML, FinTech, EdTech, and sports tech for clients in the UAE, USA, Netherlands, and Australia.
- Architected **SIMI Connect & Smart**: Stripe subscription billing (5 tiers), geo-targeted advertising, MAC-based device registry, and NestJS IoT edge firmware on Raspberry Pi with Philips Hue, Tuya, and Google Nest integrations — **20% user engagement increase**.
- Engineered **SIC Smart Interview Coach**: multi-source AI pipeline (OpenAI GPT, Whisper + ffmpeg, Hume AI, computer vision) delivering automated interview performance reports with Stripe billing and recruiter portal.
- Built **UltraHost Panel**: dual-database (MySQL + MongoDB), 4-gateway payment abstraction (Stripe, PayPal, 2Checkout, Cryptomus + webhooks), CASL RBAC, 2FA TOTP, OAuth2, and virus-scanning upload pipeline.
- Developed **Terstal AI Product Enricher**: GPT-4o-mini with Dutch retail prompt engineering, XML ERP feed ingestion, and Socket.IO real-time streaming for automated e-commerce content generation.
- Designed **shared NestJS internal boilerplate** (guards, interceptors, exception filters, role decorators) adopted across 5+ projects.

Stack: NestJS 9/10, Node.js, Vue.js 2/3, React.js 18, Next.js 14, Nuxt.js 2/3, TypeScript, MySQL, PostgreSQL, MongoDB, Redis, Firebase, TypeORM, Prisma, AWS S3, Socket.IO, MQTT, Stripe, OpenAI API, Docker, GitLab CI

One Wood Solutions
Associate Software Engineer

Lahore, Pakistan
March 2021 – January 2022

▷ Developed RESTful backend APIs with Node.js/Express.js; implemented JWT authentication, bcrypt security, MongoDB/MySQL data layers, and third-party API integrations within Agile sprint cycles.

Stack: Node.js, Express.js, MongoDB, MySQL, JavaScript, JWT, Git

PROJECT PORTFOLIO

1. DMT-ICMP/ICSP — Vehicle & Operations Management Platform

Feb 2025 – Present

Smart City · Fleet Management · Urban Cleaning · UAE

 dmt-icsp.fams.ae

▷ Enterprise platform managing fleet, workforce, maintenance, and compliance for UAE municipal Environmental Service Providers across thousands of vehicles and workers.

▷ **Core modules:** Real-time GPS tracking with Turf.js geofencing, 5-dimension schedule compliance engine, BioTime biometric attendance (5-min sync), multi-criteria PM engine (CAN bus, distance, period), 12-module inventory with 5-stage PO workflow, QEHS incident management — served via 7 client app contexts with 10+ roles and 13 automated cron jobs.

Stack: Node.js, Express.js, Vue 3, TypeScript, MySQL, Leaflet.js, Turf.js, Docker, Nginx, Oracle Cloud

2. MM-UCCP — Smart Waste & Urban Cleaning Control Platform

Feb 2025 – Present

Smart Waste Management · IoT Bin Monitoring · Tanker Dispatch · UAE

 uccp.mm.gov.qa

▷ Municipal IoT platform connecting smart bins (fill-level, RFID, temperature sensors), sewage tanker fleets, and field operators under a unified real-time control system.

▷ **Core modules:** Smart bin IoT monitoring, full tanker order lifecycle (request → dispatch → collection → discharge → completion), simulated annealing route optimization, multi-type alarm management, on-demand e-services dispatch, and 10+ report types across 4 app roles.

Stack: Node.js, Express.js, Vue 3, TypeScript, MySQL, Leaflet.js, MapLibre-GL, Firebase FCM, Docker

3. SIMI Connect & Smart — Digital Signage + IoT Smart Building Platform

Feb 2022 – Jun 2023

IoT · Digital Advertising (DOOH) · Smart Building Automation · SaaS

 simiconnect.com

▷ Multi-platform SaaS ecosystem for digital out-of-home advertising and smart building IoT automation, comprising web portals, Android apps, and edge hardware.

▷ **Core modules:** Stripe-billed group subscriptions (5 tiers, auto-renewal, grace-period), MAC-based device registry, banner campaign engine, geo-targeted advertising (location/radius/budget/time-slot), NestJS edge firmware on Raspberry Pi (Socket.IO bridge), Philips Hue mDNS integration, full Tuya Cloud API, Google Nest OAuth, MongoDB energy analytics, carbon footprint & bill estimation. **Achieved 20% user engagement increase.**

Stack: NestJS, Vue.js, PostgreSQL, MongoDB, Redis, Socket.IO, MQTT, Stripe, Auth0, Firebase, Tuya API, Philips Hue API

4. SIC — Smart Interview Coach

Sep 2023 – Jun 2024

AI SaaS · HR Technology · Interview Preparation

 smartinterviewcoach.com

▷ AI-driven SaaS delivering automated multi-dimensional interview performance analysis from 4 concurrent AI sources: OpenAI GPT (question generation + answer analysis), Whisper + ffmpeg (video-to-text), Hume AI (emotion scoring), and computer vision (posture + eye contact).

▷ **Additional:** 3 separate React portals (candidate, recruiter, admin), Stripe subscription billing with referral/coupon system, OpenAI Assistant API HR analyst chatbot, and CSV report exports.

Stack: NestJS, React.js 18, TypeScript, MySQL, OpenAI API (GPT + Whisper + Assistant), Hume AI, AssemblyAI, Stripe, AWS S3

5. UltraHost Panel — Web Hosting Billing & Management

2022 – 2024

SaaS · Web Hosting · FinTech · Multi-Gateway Billing

 ulthost.com

▷ Full-featured hosting management SaaS with 4-gateway payment abstraction (Stripe, PayPal, 2Checkout, Cryptomus — each with full webhook handling), dual-database architecture (MySQL TypeORM + MongoDB Mongoose audit logs), CASL RBAC, 2FA TOTP, OAuth2 (Google/Facebook/GitHub), and virus-scanning uploads (Cloudmersive + AWS S3).

▷ **Additional:** Invoice lifecycle cron automation, multi-currency auto rate refresh, full i18n (12+ languages), Sentry error tracking; Next.js 14 frontend with Stripe Elements, Redux Toolkit, Bootstrap 5, and jsPDF invoice generation across 25+ pages.

Stack: NestJS 10, Next.js 14, TypeScript, MySQL, MongoDB, Redis, Bull Queue, Stripe, PayPal, AWS S3, Docker, CASL, Sentry

EDUCATION

Bachelor of Science in Computer Science (BSCS)

Rawalpindi, Pakistan

PMAS-Arid Agriculture University Rawalpindi

Mar 2017 – Feb 2021 · CGPA: 3.22 / 4.00